Rotating MOG

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The abstract

I. INTRODUCTION

The intro

II. THE ROTATING MOG BLACK HOLE SURROUNDED BY QUINTESSENCE

We begin with the static, spherically symmetric metric of the MOG black hole with quintessence in Boyer-Lindquist-like coordinates (t, r, θ, ϕ),

$$ds^2 = -f(r)dt^2 + \frac{1}{f(r)}dr^2 + r^2 d\Omega^2, \tag{1}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ and the metric function $f(r)$ is defined as

$$f(r) = 1 - \frac{2(1+\alpha)Mr^2}{(r^2+\alpha(1+\alpha)M^2)^{3/2}} + \frac{\alpha(1+\alpha)M^2r^2}{(r^2+\alpha(1+\alpha)M^2)^2} - \frac{c(1+\alpha)}{r^{3w_q+1}}. \tag{2}$$

For mathematical simplicity, let us define a generalized mass function $\mathcal{M}(r)$ such that

$$f(r) = 1 - \frac{2\mathcal{M}(r)}{r}. \tag{3}$$

Multiplying through, we have

$$2\mathcal{M}(r) = r(1-f(r)) = \frac{2(1+\alpha)Mr^3}{(r^2+\alpha(1+\alpha)M^2)^{3/2}} - \frac{\alpha(1+\alpha)M^2r^3}{(r^2+\alpha(1+\alpha)M^2)^2} + \frac{c(1+\alpha)}{r^{3w_q}}. \tag{4}$$

To apply the Newman-Janis Algorithm (NJA), we first transform the coordinates from the (t, r, θ, ϕ) basis to the advanced null coordinate basis (u, r, θ, ϕ) using

$$du = dt - \frac{dr}{f(r)}. \tag{5}$$

Hence, the line element becomes

$$ds^2 = -f(r)du^2 - 2dudr + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{6}$$

We now introduce a basis of complex null tetrads $(l^\mu, n^\mu, m^\mu, \bar{m}^\mu)$ to write the contravariant metric $g^{\mu\nu}$ as

$$g^{\mu\nu} = -l^\mu n^\nu - l^\nu n^\mu + m^\mu \bar{m}^\nu + m^\nu \bar{m}^\mu. \tag{7}$$

In the (u, r, θ, ϕ) coordinate system, the orthonormal null vectors are chosen as

$$l^\mu = \delta^\mu_r \tag{8}$$

$$n^\mu = \delta^\mu_u - \frac{f(r)}{2}\delta^\mu_r \tag{9}$$

$$m^\mu = \frac{1}{\sqrt{2}r}\left(\delta^\mu_\theta + \frac{i}{\sin\theta}\delta^\mu_\phi\right) \tag{10}$$

$$\bar{m}^\mu = \frac{1}{\sqrt{2}r}\left(\delta^\mu_\theta - \frac{i}{\sin\theta}\delta^\mu_\phi\right) \tag{11}$$

These obey the standard isotropic orthogonality conditions, $l^\mu l_\mu = n^\mu n_\mu = m^\mu m_\mu = 0$ and $-l^\mu n_\mu = m^\mu \bar{m}_\mu = 1$.

The next step in the NJA is to perform the complex coordinate transformation that introduces the spin pa-

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rameter a,

$$r \to r' = r + ia\cos\theta \tag{12}$$
$$u \to u' = u - ia\cos\theta. \tag{13}$$

Under this transformation, we have that

$$r^2 \to (r+ia\cos\theta)(r-ia\cos\theta) = r^2+a^2\cos^2\theta \equiv \Sigma. \tag{14}$$

Instead of directly complexifying the generic function $f(r)$, which is mathematically ambiguous due to the r^{3w_q+1} term, we will apply Azreg-Aïnou's modified algorithm. We leave $f(r)$ real but dictate that it transforms into a new function $F(r,\theta)$ defined by

$$F(r,\theta) = \frac{f(r)r^2 + a^2\cos^2\theta}{\Sigma}. \tag{15}$$

The transformed null tetrads in the new rotating spacetime become

$$l^\mu = \delta^\mu_r \tag{16}$$
$$n^\mu = \delta^\mu_u - \frac{F}{2}\delta^\mu_r \tag{17}$$
$$m^\mu = \frac{1}{\sqrt{2\Sigma}}\left[\delta^\mu_\theta + ia\sin\theta(\delta^\mu_u - \delta^\mu_r) + \frac{i}{\sin\theta}\delta^\mu_\phi\right]. \tag{18}$$

Using these newly transformed tetrads, we obtain the contravariant components of the rotating metric in (u,r,θ,ϕ) coordinates,

$$g^{uu} = \frac{a^2\sin^2\theta}{\Sigma} \tag{19}$$
$$g^{ur} = -1 - \frac{a^2\sin^2\theta}{\Sigma} \tag{20}$$
$$g^{u\phi} = \frac{a}{\Sigma} \tag{21}$$
$$g^{rr} = F + \frac{a^2\sin^2\theta}{\Sigma} = \frac{f(r)r^2+a^2}{\Sigma} \tag{22}$$
$$g^{r\phi} = -\frac{a}{\Sigma} \tag{23}$$
$$g^{\theta\theta} = \frac{1}{\Sigma} \tag{24}$$
$$g^{\phi\phi} = \frac{1}{\Sigma\sin^2\theta}. \tag{25}$$

Inverting this tensor, we obtain the covariant components $g_{\mu\nu}$ of the metric in the same (u,r,θ,ϕ) coordinates

$$g_{uu} = -F \tag{26}$$
$$g_{ur} = -1 \tag{27}$$
$$g_{u\phi} = a(F-1)\sin^2\theta \tag{28}$$
$$g_{rr} = 0 \tag{29}$$
$$g_{r\phi} = a\sin^2\theta \tag{30}$$
$$g_{\theta\theta} = \Sigma \tag{31}$$
$$g_{\phi\phi} = \sin^2\theta\left[\Sigma + a^2(2-F)\sin^2\theta\right]. \tag{32}$$

To recover the standard form where the metric has no off-diagonal elements in r, i.e. $g_{tr}=0$ and $g_{r\phi}=0$, we will perform a final coordinate transformation back to Boyer-Lindquist coordinates (t,r,θ,ϕ_{new}). We introduce general transformation functions $\lambda(r)$ and $\chi(r)$ such that

$$\begin{cases} du &= dt - \lambda(r)dr \\ d\phi_{old} &= d\phi_{new} + \chi(r)dr \end{cases}. \tag{33}$$

Hence, the line element becomes

$$ds^2 = g_{uu}(dt-\lambda dr)^2 + 2g_{ur}(dt-\lambda dr)dr + 2g_{u\phi}(dt-\lambda dr)(d\phi_{new}+\chi dr) + 2g_{r\phi}dr(d\phi_{new}+\chi dr) + g_{\theta\theta}d\theta^2 + g_{\phi\phi}(d\phi_{new}+\chi dr)^2. \tag{34}$$

To eliminate the cross terms $dtdr$ and $drd\phi_{new}$, we gather their coefficients in (**??**) and set them to zero. The vanishing of the $dtdr$ term gives the conditions

$$-2g_{uu}\lambda + 2g_{ur} + 2g_{u\phi}\chi = 0, \tag{35}$$

which, after substituting the metric components, becomes

$$F\lambda + a(F-1)\sin^2\theta\chi = 1. \tag{36}$$

On the other hand, the vanishing of the $drd\phi_{new}$ term gives the condition

$$-2g_{u\phi}\lambda + 2g_{r\phi} + 2g_{\phi\phi}\chi = 0, \tag{37}$$

which yields

$$-a(F-1)\sin^2\theta\lambda + a\sin^2\theta + g_{\phi\phi}\chi = 0. \tag{38}$$

Now we introduce the horizon function

$$\Delta = r^2 - 2\mathcal{M}(r)r + a^2, \tag{39}$$

which gives the relation

$$F - 1 = \frac{\Delta - (r^2+a^2)}{\Sigma}. \tag{40}$$

Hence, solving the system of two linear equations, (**??**) and (**??**), for λ and χ leads to the usual transformation

functions,

$$\lambda(r) = \frac{r^2 + a^2}{\Delta} \tag{41}$$

$$\chi(r) = \frac{a}{\Delta}. \tag{42}$$

Replacing these functions in eqs. (**??**) gives the explicit global transformation required

$$\begin{cases} du & = dt - \frac{r^2+a^2}{\Delta} dr \\ d\phi_{old} & = d\phi_{new} + \frac{a}{\Delta} dr \end{cases}. \tag{43}$$

With the cross terms eliminated, the coefficient for dr^2 in eq. (**??**) gathers all remaining terms,

$$g_{rr}^{new} = g_{uu}\lambda^2 - 2g_{ur}\lambda - 2g_{u\phi}\lambda\chi + 2g_{r\phi}\chi + g_{\phi\phi}\chi^2 \tag{44}$$

which can be written in the compact form

$$g_{rr}^{new} = \frac{\Sigma}{\Delta}. \tag{45}$$

These results give the final line element for the rotating MOG black hole surrounded by a quintessence field is expressed perfectly in standard Kerr-like Boyer-Lindquist coordinates.

$$ds^2 = -\left(1 - \frac{2\mathcal{M}(r)r}{\Sigma}\right) dt^2 - \frac{4a\mathcal{M}(r)r\sin^2\theta}{\Sigma} dt d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{2\mathcal{M}(r)ra^2\sin^2\theta}{\Sigma}\right)\sin^2\theta d\phi^2. \tag{46}$$

A. Probing the rotating metric

In order to rigorously prove that the new rotating spacetime (**??**) satisfies the gravitational field equations, we can employ the orthonormal tetrad (Cartan) formalism. In generalized gravity models like MOG (Scalar-Tensor-Vector Gravity) combined with a quintessence field, applying the Newman-Janis algorithm yields a geometry that acts as a solution to the effective Einstein field equations,

$$G_{\mu\nu} = 8\pi T_{\mu\nu}^{\text{eff}}, \tag{47}$$

where the source $T_{\mu\nu}^{\text{eff}}$ is an anisotropic fluid. Therefore, the proof requires calculating the Einstein tensor of the rotating metric in a locally non-rotating frame (e.g. the Zero Angular Momentum Observer, or ZAMO, frame) and mapping it perfectly to the energy-momentum tensor of this fluid, ensuring that the physical equations of state and static limits hold. In an orthonormal frame, the energy-momentum tensor of an anisotropic fluid is diagonal:

$$T_{(b)}^{(a)} = \text{diag}(-\rho, P_r, P_t, P_t), \tag{48}$$

where ρ is the energy density, P_r is the radial pressure, and P_t is the transverse (tangential) pressure.

To find the local physical quantities measured in this locally flat frame, we project the covariant metric onto a locally flat orthonormal tetrad basis $e_\mu^{(a)}$, where the metric is Minkowski-like,

$$ds^2 = \eta_{(a)(b)} e^{(a)} e^{(b)}, \tag{49}$$

with $\eta_{(a)(b)} = \text{diag}(-1, 1, 1, 1)$. For the rotating metric (**??**), we introduce the Carter's exact orthonormal basis 1-forms, defined as

$$e^{(0)} = \sqrt{\frac{\Delta}{\Sigma}}(dt - a\sin^2\theta d\phi) \tag{50}$$

$$e^{(1)} = \sqrt{\frac{\Sigma}{\Delta}} dr \tag{51}$$

$$e^{(2)} = \sqrt{\Sigma} d\theta \tag{52}$$

$$e^{(3)} = \frac{\sin\theta}{\sqrt{\Sigma}}\left((r^2 + a^2)d\phi - a dt\right), \tag{53}$$

where Σ and Δ are given by eqs. (**??**) and (**??**), respectively. By applying Cartan's structure equations to these basis 1-forms, we compute the curvature 2-forms and ultimately the Einstein tensor $G_{(b)}^{(a)}$ in the orthonormal frame. For any stationary axisymmetric metric of this Kerr-Schild class generated by a mass function $\mathcal{M}(r)$, the non-vanishing components are rigorously established as

$$G_{(0)}^{(0)} = G_{(1)}^{(1)} = -\frac{2r^2\mathcal{M}'(r)}{\Sigma^2} \tag{54}$$

$$G_{(2)}^{(2)} = G_{(3)}^{(3)} = -\frac{r\mathcal{M}''(r)}{\Sigma} - \frac{2a^2\cos^2\theta\mathcal{M}'(r)}{\Sigma^2}. \tag{55}$$

By equating this to the source term $8\pi T_{(b)}^{(a)}$, we define the exact physical state of the effective source fluid,

$$\rho = \frac{2r^2\mathcal{M}'(r)}{8\pi\Sigma^2} \tag{56}$$

$$P_r = -\rho \tag{57}$$

$$P_t = -\frac{r\mathcal{M}''(r)}{8\pi\Sigma} - \frac{2a^2\cos^2\theta\mathcal{M}'(r)}{8\pi\Sigma^2}. \tag{58}$$

In order to prove that the metric solves the system explicitly, we must evaluate these generic pressures for the specific mass function given in eq. (**??**). First, let us

define a constant

$$K = \alpha(1+\alpha)M^2 \tag{59}$$

so that the mass function writes as

$$2\mathcal{M}(r) = \underbrace{\frac{2(1+\alpha)Mr^3}{(r^2+K)^{3/2}} - \frac{Kr^3}{(r^2+K)^2}}_{\text{MOG term}} + \underbrace{\frac{c(1+\alpha)}{r^{3w_q}}}_{\text{Quintessence term}} . \tag{60}$$

Now, we will calculate the explicit first and second radial derivatives. We have the contributions

$$\frac{d}{dr}\left(\frac{r^3}{(r^2+K)^{3/2}}\right) = \frac{3r^2(r^2+K)^{3/2} - r^3\cdot\frac{3}{2}(r^2+K)^{1/2}(2r)}{(r^2+K)^3} = \frac{3Kr^2}{(r^2+K)^{5/2}} \tag{61}$$

$$\frac{d}{dr}\left(\frac{r^3}{(r^2+K)^2}\right) = \frac{3r^2(r^2+K)^2 - r^3\cdot 2(r^2+K)(2r)}{(r^2+K)^4} = \frac{3Kr^2 - r^4}{(r^2+K)^3}. \tag{62}$$

These terms give the first derivative of the mass function

$$2\mathcal{M}'(r) = 2(1+\alpha)M\left[\frac{3Kr^2}{(r^2+K)^{5/2}}\right] - K\left[\frac{3Kr^2 - r^4}{(r^2+K)^3}\right] - \frac{3w_q c(1+\alpha)}{r^{3w_q+1}}. \tag{63}$$

Similarly, for the second derivative we have the terms

$$\frac{d}{dr}\left(\frac{3Kr^2}{(r^2+K)^{5/2}}\right) = \frac{6Kr(r^2+K)^{5/2} - 3Kr^2\cdot\frac{5}{2}(r^2+K)^{3/2}(2r)}{(r^2+K)^5} = \frac{3K(2Kr-3r^3)}{(r^2+K)^{7/2}} \tag{64}$$

$$\frac{d}{dr}\left(\frac{3Kr^2 - r^4}{(r^2+K)^3}\right) = \frac{(6Kr-4r^3)(r^2+K)^3 - (3Kr^2-r^4)\cdot 3(r^2+K)^2(2r)}{(r^2+K)^6} = \frac{6K^2r - 16Kr^3 + 2r^5}{(r^2+K)^4} \tag{65}$$

$$\frac{d}{dr}\left(\frac{-3w_q c(1+\alpha)}{r^{3w_q+1}}\right) = \frac{3w_q(3w_q+1)c(1+\alpha)}{r^{3w_q+2}}, \tag{66}$$

which give the result

$$2\mathcal{M}''(r) = 2(1+\alpha)M\left[\frac{3K(2Kr-3r^3)}{(r^2+K)^{7/2}}\right] - K\left[\frac{6K^2r - 16Kr^3 + 2r^5}{(r^2+K)^4}\right] + \frac{3w_q(3w_q+1)c(1+\alpha)}{r^{3w_q+2}}. \tag{67}$$

Substituting these derivatives into equations (**??**) and (**??**), we obtain the energy density ρ and tangential pressure P_t for this spacetime. Meanwhile, equation (**??**) is an equation of state that gives the form of the radial pressure as $P_r = -\rho$, i.e. with a coefficient $\kappa_{radial} = -1$, which is a defining characteristic of the energy-momentum tensor of a dark energy (quintessence) field and regular modified gravity models. This guarantees that the geometry remains regular without naked singularities.

Now, if we consider the asymptotic behavior, i.e. large r, and ignore the MOG vector field ($M \to 0, K \to 0$), we

isolate the quintessence field. There, we have

$$2\mathcal{M}'(r) \approx \frac{-3w_q c(1+\alpha)}{r^{3w_q+1}} \tag{68}$$

and therefore,

$$\rho \propto \frac{2r^2}{\Sigma^2}\left(\frac{-w_q}{r^{3w_q+1}}\right). \tag{69}$$

Since w_q is negative (remember that $-1 < w_q < -1/3$), it is clear that the local energy density of the quintessence field ρ is strictly positive ($\rho > 0$), satisfying the weak energy condition.

Finally, we analyse the static limit recovery, which is obtained considering that rotation ceases, i.e. $a \to 0$, which implies that the source fluid reduces to the source of the original seed metric (**??**). Hence, note that considering $a = 0$ gives $\Sigma = r^2$ and the tangential pressure limits to

$$\lim_{a\to 0} P_t = \lim_{a\to 0}\left(-\frac{r\mathcal{M}''(r)}{\Sigma} - \frac{2a^2\cos^2\theta\mathcal{M}'(r)}{\Sigma^2}\right) = -\frac{\mathcal{M}''(r)}{r}, \tag{70}$$

which cleanly recovers $G^\theta_\theta = G^\phi_\phi = -\mathcal{M}''(r)/r$, and perfectly matches the static, spherically symmetric field equations for MOG and Quintessence.

B. Energy Conditions

The physical viability of the effective anisotropic fluid sourcing the rotating MOG-quintessence spacetime is assessed through the standard pointwise energy conditions [**?**]. Using the orthonormal frame components (ϵ, P_r, P_t) derived in eqs. (**??**)–(**??**), the relevant conditions are:

- *Weak Energy Condition* (WEC): $\epsilon \geq 0$ and $\epsilon + p_i \geq 0$ for each pressure component p_i.
- *Strong Energy Condition* (SEC): $\epsilon + \sum_i p_i \geq 0$ and $\epsilon + p_i \geq 0$.
- *Dominant Energy Condition* (DEC): $\epsilon \geq |p_i|$.

Since the equation of state (**??**) gives $P_r = -\rho$ exactly, the combination $\epsilon + P_r = 0$ is trivially satisfied on its boundary, while the non-trivial constraints come from the tangential pressure P_t. Figure **??** displays the four independent combinations as functions of r/M for the equatorial plane ($\theta = \pi/2$), with $w_q = -2/3$, $a = 0.5$, and $M = 1$.

C. Horizons

The event horizons of the black hole correspond to the coordinate singularities of the metric, specifically where the radial component of the contravariant metric vanishes, i.e. $g^{rr} = 0$. In Boyer-Lindquist coordinates, this condition is exactly equivalent to finding the roots of the horizon function,

$$\Delta = r^2 - 2\mathcal{M}(r)r + a^2 = 0. \tag{71}$$

Substituting the explicit form of the generalized mass function (**??**), this equation becomes

$$r^2 - \left(\frac{2(1+\alpha)Mr^3}{(r^2+\alpha(1+\alpha)M^2)^{3/2}} - \frac{\alpha(1+\alpha)M^2r^3}{(r^2+\alpha(1+\alpha)M^2)^2} + \frac{c(1+\alpha)}{r^{3w_q}}\right) + a^2 = 0. \tag{72}$$

Depending on the parameters α, a, c, and w_q, this equation may admit multiple roots, a single degenerate root (extremal black hole), or no real roots (naked singularity).

The interplay between these parameters controls the entire causal structure of the spacetime. Figure **??** illustrates how the horizon function $\Delta(r)$ responds to variations in α, a, and c, while Figs. **??** and **??** show how the horizon radii and the extremal spin depend on the physical parameters. A representative sample of horizon locations for selected parameter combinations is given in Table **??**.

D. The Static Limit Surface and the Ergosphere

The static limit surface (SLS) is the boundary where the asymptotic time-translation Killing vector $\xi^\mu = \partial_t$ becomes null. Physically, this represents the threshold where an observer can no longer remain stationary relative to spatial infinity and is inevitably forced to co-rotate with the black hole due to frame dragging. This boundary is found by setting the condition

$$g_{tt} = -\left(1 - \frac{2\mathcal{M}(r)r}{\Sigma}\right) = 0 \tag{73}$$

which yields

$$\Sigma - 2\mathcal{M}(r)r = 0. \tag{74}$$

FIG. 1. Energy conditions for the effective anisotropic fluid of the rotating MOG black hole surrounded by quintessence, evaluated on the equatorial plane with $w_q = -2/3$, $a = 0.5$, and $M = 1$. Vertical dotted lines indicate the positions of the horizons for each parameter set. *Top-left:* Weak energy condition, tangential component $\epsilon + p_\phi \geq 0$, for four values of α with $c = 0.01$. The combination remains non-negative throughout the exterior region, confirming WEC satisfaction. The MOG parameter α amplifies the energy-density peak near the black hole: the stronger gravitational coupling of STVG increases the effective source density at small radii, producing a pronounced maximum that shifts inward as α grows. *Top-right:* Strong energy condition trace $\epsilon + \sum_i p_i$ for the same values of α. This quantity is *violated* (< 0) in the region between the Cauchy and event horizons for all $\alpha > 0$, and the violation deepens markedly with increasing α. The SEC failure is a direct consequence of the dark-energy equation of state $P_r = -\epsilon$: the combined MOG–quintessence source mimics a cosmological-constant-like fluid near the singularity, which is known to violate the SEC. For $\alpha = 0$ (pure GR with quintessence), the SEC is marginally satisfied, underscoring that the violation is driven by the MOG sector. *Bottom-left:* Dominant energy condition $\epsilon - |p_\phi| \geq 0$ as a function of the quintessence normalization c with $\alpha = 0.3$. All curves are positive and nearly degenerate, showing that the DEC is robustly satisfied and is essentially insensitive to the quintessence parameter c: the dark-energy background shifts the horizon positions (vertical dotted lines) but does not alter the local energy hierarchy between ϵ and $|p_\phi|$. *Bottom-right:* Same DEC combination for varying α with $c = 0.01$. The DEC is satisfied for all values of α, with the peak broadening slightly as α increases due to the widening of the effective MOG potential well. Together, these panels establish that the rotating MOG-quintessence spacetime satisfies the WEC and DEC everywhere in the exterior region, while the SEC is violated near the horizon — a hallmark of dark-energy-dominated matter that is entirely consistent with the quintessence equation of state $-1 < w_q < -1/3$.

FIG. 2. Qualitative behavior of the horizon function $\Delta(r)$ for the rotating MOG black hole surrounded by quintessence, with $w_q = -2/3$ and $M = 1$. *Left panel:* Effect of the MOG coupling parameter α with fixed spin $a = 0.5$ and quintessence normalization $c = 0.01$. As α increases the gravitational well deepens and the event horizon r_+ shifts outward: the stronger gravitational attraction of MOG pulls the horizon further from the singularity. A cosmological root r_∞ (where Δ crosses zero from above at large r) is present for all cases shown and moves inward as α grows. *Right panel:* Effect of the spin parameter a with $\alpha = 0.3$ and $c = 0.01$. Increasing a raises the minimum of Δ; the Cauchy horizon r_- and the event horizon r_+ approach each other until they merge at the extremal value a_E, beyond which no horizon exists and the singularity becomes naked. The cosmological root is nearly unaffected by a. In all cases $\Delta \to -\infty$ as $r \to 0^+$, confirming the ring singularity, and $\Delta \to -\infty$ for $r \to \infty$ due to the quintessence term $-c(1+\alpha)/r^{3w_q}$ with $w_q < -1/3$.

Recalling that $\Sigma = r^2 + a^2\cos^2\theta$, the condition for the static limit surface is

$$r^2 - 2\mathcal{M}(r)r + a^2\cos^2\theta = 0 \tag{75}$$

Substituting the mass function (**??**) gives

$$r^2 - \left(\frac{2(1+\alpha)Mr^3}{(r^2+\alpha(1+\alpha)M^2)^{3/2}} - \frac{\alpha(1+\alpha)M^2r^3}{(r^2+\alpha(1+\alpha)M^2)^2} + \frac{c(1+\alpha)}{r^{3w_q}}\right) + a^2\cos^2\theta = 0. \tag{76}$$

The ergosphere is the topologically toroidal region bounded between the outer event horizon and the outer static limit surface ($r_+ < r < r_{SLS}$). Because $g_{tt} > 0$ within this region, the temporal coordinate t becomes space-like, meaning that particles are mandated to move in the direction of the black hole's rotation.

Two properties of the SLS follow directly from its defining equation. At the poles ($\theta = 0, \pi$), the condition $r^2 + a^2 - 2\mathcal{M}(r)r = 0$ is identical to $\Delta = 0$, so $r_{SLS}(\theta{=}0) = r_+$ and the ergosphere has zero thickness there. At the equator ($\theta = \pi/2$), the $a^2\cos^2\theta$ term vanishes and the SLS condition reduces to

$$r^2 - 2\mathcal{M}(r)r = 0, \tag{77}$$

which is *independent of the spin parameter* a. Consequently, the equatorial SLS radius $r_{SLS}(\pi/2)$ is fixed by the mass profile alone, while the event horizon r_+ shrinks as a increases toward the extremal value $a_E(\alpha, c)$. The equatorial ergosphere width therefore grows monotonically with a,

$$\Delta r_{\rm ergo} \equiv r_{SLS}(\pi/2) - r_+(\alpha, a, c) \xrightarrow{a \to a_E} r_{SLS}(\pi/2) - r_E, \tag{78}$$

where r_E is the degenerate extremal horizon radius.

Figure **??** shows meridional cross-sections of the ergosphere (shaded region between solid SLS and dashed horizon lines) for representative parameter values. In panel (a), the spin is held fixed at $a = 0.40\,M$ while α varies. Larger values of the MOG parameter place a progressively closer to $a_E(\alpha)$, which decreases with α (see

FIG. 3. Horizon radii as functions of the quintessence normalization c, the MOG parameter α, and the spin a, for $w_q = -2/3$ and $M = 1$. Solid lines denote the event horizon r_+, dashed lines the Cauchy horizon r_-, and dotted lines the cosmological horizon r_∞. *Panel (a):* Dependence on c for fixed $a = 0.5$ and three values of α. As c grows, the quintessence contribution to $\mathcal{M}(r)$ increases, driving r_+ upward and r_∞ downward; the two horizons eventually merge at a critical c marking the transition to a cosmological extremal configuration. Larger α raises r_+ for all c because the MOG-enhanced gravitational coupling effectively increases the black hole mass felt at intermediate radii. *Panel (b):* Dependence on α for fixed $a = 0.5$ and $c = 0.01$. The event horizon grows monotonically with α while the Cauchy horizon shrinks slightly, consistent with the MOG correction fattening the gravitational potential. The cosmological horizon decreases with α because the same quintessence normalization c is multiplied by $(1 + \alpha)$ in the mass function, effectively strengthening the dark-energy contribution. *Panel (c):* Dependence on a for fixed $c = 0.01$ and four values of α. For each α the two black-hole horizons $r_\pm$ converge as $a \to a_E(\alpha)$, while r_∞ is essentially independent of a. Larger α shifts a_E to higher values because the deeper gravitational potential requires more centrifugal repulsion to reach the extremal condition.

FIG. 4. Extremal spin parameter a_E — the maximum spin for which a black hole horizon exists — as a function of the quintessence normalization c and the MOG coupling α, for $w_q = -2/3$ and $M = 1$. *Left panel:* a_E vs. c for four values of α. Increasing c reduces a_E: a stronger quintessence background deepens the effective potential at large r, narrowing the window of spin values that admit an event horizon. Higher α lifts the entire $a_E(c)$ curve, because the stronger MOG gravity counteracts the spin-induced centrifugal barrier more effectively. *Right panel:* a_E vs. α for four values of c. The extremal spin grows roughly linearly with α at fixed c; this is a direct consequence of the MOG gravitational enhancement, which allows the black hole to sustain a larger angular momentum while still possessing a well-defined horizon. A larger c uniformly depresses the curve, confirming that quintessence always acts to destabilize the horizon structure by reinforcing the effective dark-energy repulsion at intermediate radii. In the limit $c \to 0$, all curves converge to the standard rotating MOG result without quintessence.

Fig. **??**), resulting in a wider and more oblate ergosphere. In panel (b), $\alpha = 0.3$ is fixed while a increases from $0.15\,M$ to $0.72\,M$: the outer SLS boundary remains unchanged (consistent with its a-independence at the equator), whereas the inner horizon boundary contracts, enlarging the ergosphere region.

The quantitative growth of the ergosphere with spin is displayed in Fig. **??**, which plots $\Delta r_{\rm ergo}$ as a function of a/M for four values of α at fixed $c = 0.01$. Each curve terminates at the corresponding extremal spin $a_E(\alpha)$: the divergence-like rise near termination reflects the collapse of $r_+ \to r_E$ while $r_{SLS}(\pi/2)$ remains finite. A key result is that increasing α *decreases* a_E, shifting the terminal point of each curve to smaller values of a/M. Thus, for any fixed spin, a more strongly modified-gravity spacetime (α larger) supports a larger ergosphere and therefore a broader region where energy extraction via the Penrose process is kinematically permitted.

III. MOTION OF MASSIVE TEST PARTICLES

We consider a particle of rest mass $\mu = 1$ moving in the surroundings of the rotating MOG black hole (**??**). The geodesic equations can be elegantly obtained by using the Hamilton-Jacobi equation,

$$\frac{\partial S}{\partial \tau} + \frac{1}{2} g^{\mu\nu} \frac{\partial S}{\partial x^\mu} \frac{\partial S}{\partial x^\nu} = 0 \tag{79}$$

where S is the Jacobi action and τ is the proper time. Because the rotating metric is stationary ($\partial_t g_{\mu\nu} = 0$) and axisymmetric ($\partial_\phi g_{\mu\nu} = 0$), the energy E and the axial angular momentum L of the particle are conserved. Furthermore, the Kerr-like structure of the metric admits a Killing tensor, guaranteeing the separability of the Hamilton-Jacobi equation and yielding a third constant of motion, the Carter constant $\mathcal{Q}$. We apply the standard separable ansatz for the action,

$$S = -Et + L\phi + S_r(r) + S_\theta(\theta), \tag{80}$$

to substitute into the Hamilton-Jacobi equation and utilizing the contravariant metric components $g^{\mu\nu}$, the system completely separates into independent radial and angular components.

By applying the relation $g_{\mu\nu} \frac{dx^\nu}{d\tau} = \partial_\mu S$, we derive the four first-order differential equations in Boyer-Lindquist coordinates,

$$\Sigma \frac{dt}{d\tau} = a(L - aE\sin^2\theta) + \frac{r^2 + a^2}{\Delta}[E(r^2 + a^2) - aL] \tag{81}$$

$$\Sigma \frac{dr}{d\tau} = \pm\sqrt{V_r} \tag{82}$$

$$\Sigma \frac{d\theta}{d\tau} = \pm\sqrt{V_\theta} \tag{83}$$

$$\Sigma \frac{d\phi}{d\tau} = \frac{L}{\sin^2\theta} - aE + \frac{a}{\Delta}[E(r^2 + a^2) - aL]. \tag{84}$$

A. Effective Potential for Massive Particles

For motion confined to the equatorial plane ($\theta = \pi/2$), we set the Carter constant $\mathcal{Q} = 0$ and the radial potential V_r then simplifies to

$$V_r = [E(r^2 + a^2) - aL]^2 - \Delta[r^2 + (L - aE)^2]. \tag{85}$$

FIG. 5. Meridional cross-sections of the ergosphere in the (x, z) plane, where $x = r\sin\theta$ and $z = r\cos\theta$. Solid lines denote the static limit surface; dashed lines denote the event horizon r_+. The ergosphere occupies the shaded band between them. *Panel (a):* Fixed spin $a = 0.40\,M$ and $c = 0.01$, varying MOG parameter $\alpha \in \{0, 0.2, 0.4, 0.6\}$. Increasing α reduces $a_E(\alpha)$, placing the fixed spin closer to extremality and producing a larger, more oblate ergosphere. *Panel (b):* Fixed $\alpha = 0.3$, $c = 0.01$, varying spin $a \in \{0.15, 0.40, 0.60, 0.72\}\,M$ (all below the extremal value $a_E \approx 0.76\,M$). The equatorial SLS radius is constant, while the horizon shrinks as a grows, widening the ergosphere. All lengths in units of GM/c^2.

TABLE I. Numerical values of the Cauchy horizon r_-, event horizon r_+, and cosmological horizon r_∞ for selected combinations of the spin a, MOG parameter α, and quintessence normalization c, with $w_q = -2/3$ and $M = 1$. A dash (—) indicates that the corresponding root does not exist for the given parameter set. The extremal condition $r_- = r_+$ is reached as $a \to a_E(\alpha, c)$. All radii are in units of the gravitational radius GM/c^2.

α	a	c	r_-	r_+	r_∞
0.0	0.0	0.01	—	2.000	46.42
0.0	0.3	0.01	0.154	1.983	46.42
0.0	0.5	0.01	0.268	1.939	46.42
0.0	0.8	0.01	0.545	1.789	46.42
0.0	1.0	0.01	1.000	1.000	46.42
0.3	0.0	0.01	—	2.292	43.17
0.3	0.3	0.01	0.135	2.274	43.17
0.3	0.5	0.01	0.237	2.230	43.17
0.3	0.8	0.01	0.501	2.077	43.17
0.6	0.0	0.01	—	2.554	40.37
0.6	0.5	0.01	0.213	2.506	40.37
0.6	0.8	0.01	0.461	2.352	40.37
1.0	0.0	0.01	—	2.883	37.21
1.0	0.5	0.01	0.184	2.835	37.21
1.0	0.8	0.01	0.412	2.681	37.21
0.3	0.5	0.00	0.226	2.237	—
0.3	0.5	0.02	0.241	2.226	28.15
0.3	0.5	0.05	0.247	2.222	19.31
0.3	0.5	0.08	0.253	2.218	14.40
0.3	0.5	0.10	0.257	2.216	12.30

FIG. 6. Equatorial ergosphere width $\Delta r_{\rm ergo} = r_{SLS}(\pi/2) - r_+$ as a function of the dimensionless spin a/M for $\alpha \in \{0, 0.2, 0.4, 0.6\}$ at fixed $c = 0.01$. Each curve terminates at the extremal spin $a_E(\alpha)$, which decreases with α as shown in Fig. **??**. The divergence near termination reflects the coalescence of the Cauchy and event horizons at extremality. The monotonic growth of $\Delta r_{\rm ergo}$ with a is a consequence of the a-independence of $r_{SLS}(\pi/2)$ combined with the reduction of r_+.

The effective potential V_{eff} is defined such that the radial motion satisfies $\left(\frac{dr}{d\tau}\right)^2 + V_{eff} = 0$. For equatorial orbits, this is derived from the condition $V_r = 0$,

$$V_{eff}(r) = -\frac{\left[E(r^2+a^2) - aL\right]^2 - \Delta[r^2 + (L-aE)^2]}{r^4}. \tag{86}$$

Figure **??** illustrates the structure of V_r/r^4 for prograde equatorial orbits. Panel (a) contrasts an unstable orbit located below the ISCO with three stable orbits above it: for the unstable orbit V_r has a local maximum at $r_{\rm circ}$, whereas for the stable ones it presents a minimum there, with the ISCO corresponding to the inflection point where both $V_r = 0$ and $V_r'' = 0$ simultaneously. Panel (b) shows how the inflection point shifts outward as the MOG parameter α increases, with the filled circles marking $r_{\rm ISCO}(\alpha)$: larger α deepens and widens the potential well, pushing the innermost stable orbit to larger radii.

B. Innermost Stable Circular Orbit (ISCO)

Circular orbits occur when the radial velocity and acceleration both vanish, i.e.

$$V_r(r) = 0, \quad \frac{dV_r}{dr} = 0 \tag{87}$$

For equatorial circular motion around the rotating spacetime (**??**) with the variable mass function $\mathcal{M}(r)$, these conditions give the energy E and angular momentum L as

$$E = \frac{r^2 - 2\mathcal{M}(r)r \pm a\sqrt{\Phi(r)}}{r\sqrt{r^2 - 3\mathcal{M}(r)r \pm 2a\sqrt{\Phi(r)}}} \tag{88}$$

and

$$L = \pm\frac{\sqrt{\Phi(r)}\left(r^2 + a^2 \mp 2a\sqrt{\Phi(r)}\right)}{r\sqrt{r^2 - 3\mathcal{M}(r)r \pm 2a\sqrt{\Phi(r)}}} \tag{89}$$

where we define the auxiliary function $\Phi(r) = \mathcal{M}(r)r + r^2\mathcal{M}'(r)$. Considering the MOG-quintessence mass function (**??**) and defining $K = \alpha(1+\alpha)M^2$, we have

$$2\mathcal{M}(r) = \frac{2(1+\alpha)Mr^3}{(r^2+K)^{3/2}} - \frac{Kr^3}{(r^2+K)^2} + \frac{c(1+\alpha)}{r^{3w_q}} \tag{90}$$

and its derivative

$$2\mathcal{M}'(r) = \frac{6K(1+\alpha)Mr^2}{(r^2+K)^{5/2}} - \frac{K(3Kr^2 - r^4)}{(r^2+K)^3} - \frac{3w_q c(1+\alpha)}{r^{3w_q+1}}. \tag{91}$$

Hence, the auxiliary function $\Phi(r)$ simplifies to

FIG. 7. Effective radial potential V_r/r^4 for prograde equatorial orbits with $a = 0.5M$, $c = 0.005$, $w_q = -2/3$. *Left:* V_r/r^4 evaluated at the conserved (E, L) corresponding to circular orbits of radius $r_{\rm circ}$, with $\alpha = 0.3$. The dotted blue curve (orbit below the ISCO) has a local maximum at $r_{\rm circ}$, signalling orbital instability; solid curves (orbits above the ISCO) display a local minimum, confirming stability. The dashed vertical line marks $r_{\rm ISCO} = 4.90M$, the inflection point where both $V_r = V_r' = V_r'' = 0$. *Right:* V_r/r^4 evaluated at the ISCO energy and angular momentum for $\alpha \in \{0, 0.2, 0.4, 0.5\}$. Filled circles mark the corresponding $r_{\rm ISCO}(\alpha)$; increasing α shifts the potential well outward, indicating that the MOG vector field effectively weakens the gravitational binding and enlarges the ISCO.

$$\Phi(r) = \frac{(1+\alpha)Mr^4(r^2+4K)}{(r^2+K)^{5/2}} - \frac{2K^2r^4}{(r^2+K)^3} + \frac{c(1+\alpha)(1-3w_q)}{2r^{3w_q-1}} \tag{92}$$

C. ISCO

The stability of these circular orbits is determined by the second derivative of the potential. The ISCO radius, r_{ISCO}, is the smallest radius for which a stable circular orbit can exist. By explicitly evaluating $\partial_{rr}V_r = 0$ at the location of a circular orbit, we arrive at the following characteristic equation for r_{ISCO}

$$r^2 - 6\mathcal{M}(r)r \pm 8a\sqrt{\Phi(r)} - 3a^2 + \Psi(r) = 0, \tag{93}$$

where $\Psi(r)$ is the correction function due to the non-constancy of the MOG and quintessence mass profile,

$$\Psi(r) = 3r^2\mathcal{M}'(r) - \frac{r^3\mathcal{M}''(r)[r^2 - 3\mathcal{M}(r)r \pm 2a\sqrt{\Phi(r)}]}{\Phi(r)}. \tag{94}$$

Solving this transcendental equation for r yields the radius of the Innermost Stable Circular Orbit. In the limit of General Relativity ($\mathcal{M}' = \mathcal{M}'' = 0$), this equation reduces to the standard Kerr ISCO condition: $r^2 - 6Mr \pm 8a\sqrt{Mr} - 3a^2 = 0$.

The numerical results are presented in Figs. **??** and **??**. Figure **??** displays $r_{\rm ISCO}$ as a function of spin a for both prograde (solid) and retrograde (dashed) orbits. In panel (a), increasing α at fixed c shifts both branches outward: the MOG gravitational coupling effectively increases the total mass felt by the test particle, requiring a larger centrifugal barrier to maintain a stable circular orbit. Additionally, larger α reduces the extremal spin a_E at which the curves terminate, so the allowed phase space for stable orbits narrows. Panel (b) demonstrates that increasing the quintessence normalisation c amplifies this effect: the cosmological dark-energy background acts as a repulsive term that further destabilises inner orbits, raising $r_{\rm ISCO}$ and compressing the accessible spin range.

Figure **??** shows the ISCO-specific energy $E_{\rm ISCO}$ and angular momentum $L_{\rm ISCO}$ as functions of α for several prograde spin values. Since the radiative efficiency of a thin accretion disc is $\eta = 1 - E_{\rm ISCO}$, panel (a) directly encodes the disc luminosity: $E_{\rm ISCO}$ increases monotonically with α, meaning MOG *reduces* the maximum radiative efficiency relative to Kerr. This weakening of binding energy is a direct consequence of the repulsive contribution of the MOG vector field. Panel (b) confirms the complementary picture: $L_{\rm ISCO}$ also increases with α, so infalling matter carries greater angular momentum to the horizon, broadening the innermost disc annulus. Together, these results suggest that observations of the disc emission and quasi-periodic oscillations of X-ray binaries or active galactic nuclei could potentially constrain α independently of the spin measurement.

IV. PHOTON DYNAMICS AND SHADOW OF THE ROTATING MOG-QUINTESSENCE BLACK HOLE

To understand the optical appearance of the rotating MOG-Quintessence black hole, we must analyze the trajectories of photons (null geodesics) traveling through this spacetime. The geodesic equations can be elegantly obtained by using the Hamilton-Jacobi equation. By taking the derivatives of the action with respect to the conserved quantities, we obtain the four first-order differential Hamiltonian equations of motion governing the trajectory of the photon,

$$\Sigma\frac{dt}{d\lambda} = \frac{r^2+a^2}{\Delta}\left[E(r^2+a^2) - aL\right] - a(aE\sin^2\theta - L) \tag{95}$$

$$\Sigma\frac{dr}{d\lambda} = \pm\sqrt{\mathcal{R}(r)} \tag{96}$$

$$\Sigma\frac{d\theta}{d\lambda} = \pm\sqrt{\Theta(\theta)} \tag{97}$$

$$\Sigma\frac{d\phi}{d\lambda} = \frac{a}{\Delta}\left[E(r^2+a^2) - aL\right] - \left(aE - \frac{L}{\sin^2\theta}\right), \tag{98}$$

where λ is an affine parameter, while the functions $\mathcal{R}(r)$ and $\Theta(\theta)$ are the radial and angular effective potentials, respectively. Note that the photon's motion is strictly constrained to regions where $\mathcal{R}(r) \geq 0$ and $\Theta(\theta) \geq 0$. In order to write explicitly these effective potentials, we define the impact parameters (conserved quantities normalized by energy) as

$$\xi = \frac{L}{E}, \quad \eta = \frac{\mathcal{Q}}{E^2}. \tag{99}$$

FIG. 8. ISCO radius $r_{\rm ISCO}$ as a function of spin a/M for prograde (solid) and retrograde (dashed) equatorial orbits. *Left:* Effect of the MOG parameter α at fixed $c = 0.01$ and $w_q = -2/3$. Each pair of curves terminates at the extremal-BH limit $a_E(\alpha)$, which decreases with α. In the Schwarzschild limit ($a = 0$) the prograde and retrograde ISCOs are degenerate and both shift outward with α, while for $a > 0$ the prograde ISCO is always interior to the retrograde one. *Right:* Effect of the quintessence normalisation c at fixed $\alpha = 0.3$ and $w_q = -2/3$. The prograde curves are nearly coincident, indicating that quintessence has a subdominant effect on the prograde ISCO at moderate spins; the retrograde ISCO shows a slightly stronger response to c, and all curves terminate at a reduced $a_E(c)$.

FIG. 9. Specific energy $E_{\rm ISCO}/\mu$ (left) and angular momentum $L_{\rm ISCO}/(\mu M)$ (right) of prograde ISCO orbits as functions of the MOG parameter α, for $c = 0.01$ and $w_q = -2/3$. Each curve terminates at the extremal-BH limit $\alpha_E(a)$. The radiative efficiency of a geometrically thin, optically thick accretion disc is $\eta = 1 - E_{\rm ISCO}$; since $E_{\rm ISCO}$ increases monotonically with α, the MOG field systematically suppresses the disc luminosity relative to the Kerr case. The angular momentum $L_{\rm ISCO}$ grows with α for all spins, indicating that the innermost disc ring carries progressively more angular momentum as the MOG coupling strengthens.

Then, using these impact parameters, the effective radial potential is

$$\frac{\mathcal{R}(r)}{E^2} = (r^2 + a^2 - a\xi)^2 - \Delta\left[\eta + (\xi - a)^2\right], \tag{100}$$

and the effective angular potential is

$$\frac{\Theta(\theta)}{E^2} = \eta + a^2\cos^2\theta - \xi^2\cot^2\theta. \tag{101}$$

The roots of $\mathcal{R}(r) = 0$ determine the radial turning points of the photon's trajectory, while the roots of $\Theta(\theta) = 0$ define the latitudinal boundaries of the orbit.

The photon sphere is a critical region where photons can orbit the black hole in closed, unstable circular paths. These orbits dictate the boundary between photons that fall into the black hole and those that scatter back to infinity, effectively forming the edge of the black hole's shadow. The conditions for an unstable circular photon orbit of radius r_p are

$$\mathcal{R}(r_p) = 0 \tag{102}$$
$$\mathcal{R}'(r_p) = 0 \tag{103}$$
$$\mathcal{R}''(r_p) > 0. \tag{104}$$

Applying the first two conditions to our effective radial potential yields a system of two equations

$$\begin{cases} (r_p^2 + a^2 - a\xi)^2 - \Delta(r_p)\left[\eta + (\xi - a)^2\right] = & 0 \\ 4r_p(r_p^2 + a^2 - a\xi) - \Delta'(r_p)\left[\eta + (\xi - a)^2\right] = & 0 \end{cases}. \tag{105}$$

By dividing the second equation by the first, we can solve exactly for the critical impact parameters ξ and η as functions of the photon orbit radius r_p,

$$\xi(r_p) = \frac{r_p^2 + a^2}{a} - \frac{4r_p\Delta(r_p)}{a\Delta'(r_p)} \tag{106}$$

$$\eta(r_p) = \frac{16r_p^2\Delta(r_p)}{(\Delta'(r_p))^2} - \frac{1}{a^2}\left(r_p^2 - \frac{4r_p\Delta(r_p)}{\Delta'(r_p)}\right)^2. \tag{107}$$

Because of the rotation parameter a, the photon sphere has a radius that range from a minimum prograde orbit (r_p^-) to a maximum retrograde orbit (r_p^+), depending on whether the photon revolves with or against the black hole's spin.

The shadow of the black hole is the optical silhouette cast by the compact object against a bright background, bounded by the locus of the unstable circular photon orbits. To describe the shadow's geometry as seen by a distant observer located at a coordinate distance $r_o \to \infty$ and an inclination angle θ_o, we project the photon trajectories onto the observer's local celestial sky. In order to do this, we introduce the standard celestial coordinates (α, β), which represent the apparent perpendicular displacement of the shadow on the observer's viewing screen,

$$\alpha = \lim_{r_o\to\infty}\left(-r_o^2\sin\theta_o\frac{d\phi}{dr}\right) \tag{108}$$

$$\beta = \lim_{r_o\to\infty}\left(r_o^2\frac{d\theta}{dr}\right). \tag{109}$$

Substituting the Hamiltonian equations of motion $\frac{d\phi}{dr} = \frac{\dot\phi}{\dot r}$ and $\frac{d\theta}{dr} = \frac{\dot\theta}{\dot r}$ evaluated at spatial infinity, the celestial coordinates reduce to pure algebraic expressions of the critical impact parameters,

$$\alpha(r_p) = -\frac{\xi(r_p)}{\sin\theta_o} \tag{110}$$

$$\beta(r_p) = \pm\sqrt{\eta(r_p) + a^2\cos^2\theta_o - \xi(r_p)^2\cot^2\theta_o}. \tag{111}$$

For the most common configuration where the observer is situated on the equatorial plane ($\theta_o = \pi/2$), and then the celestial coordinates simplify to

$$\alpha = -\xi(r_p) \tag{112}$$

$$\beta = \pm\sqrt{\eta(r_p)} \tag{113}$$

By sweeping the parameter r_p continuously between the prograde and retrograde orbit radii, the parametric curve

$(\alpha(r_p), \beta(r_p))$ plots the exact closed boundary of the black hole's shadow. The rotating MOG metric (**??**), includes many parameters and therefore the exact form and size of the shadow will depend on the values of all of them. In general, the behavior is such that the MOG parameter α increases the gravitational pull, thereby widening the overall radius of the shadow. Meanwhile, the Quintessence parameter c warps the background spacetime, distorting the shadow's area depending on the negative equation of state w_q. Finally, the Spin parameter a introduces an asymmetry, flattening one side of the shadow (the "D-shape" observed by the Event Horizon Telescope) due to frame-dragging actively pulling the prograde photons closer to the event horizon.

Figure **??** illustrates these three effects independently. Panel (a) shows that increasing α enlarges the shadow while preserving a nearly circular outline, since the MOG gravitational coupling acts isotropically. Panel (b) shows that the quintessence normalization c similarly enlarges the shadow without modifying its shape, because the quintessence field contributes a spherically symmetric effective mass term. Panel (c) demonstrates the qualitatively distinct effect of spin: as a increases from zero, the shadow develops a pronounced D-shape, with the prograde side flattening toward the event horizon due to frame-dragging, while the retrograde side bulges outward. The spin range shown is limited by the extremal condition $a \leq a_E(\alpha, c)$: as established in Fig. **??**, for $\alpha = 0.3$ and $c = 0.005$ the maximum spin that admits a well-defined event horizon is $a_E \approx 0.75$, so the largest value displayed ($a = 0.7$) lies close to the extremal limit where the two horizons are nearly degenerate.

The three effects are not independent observables: while α and c each set the overall scale of the shadow, only the spin a controls its asymmetry. This separation is made explicit by Figure **??**, which overlays the shadow boundaries for the Kerr metric ($\alpha = 0$, $c = 0$), rotating MOG without quintessence ($c = 0$), and the full rotating MOG-quintessence geometry. The hierarchy

$$R_s^{\mathrm{Kerr}} < R_s^{\mathrm{MOG}} < R_s^{\mathrm{MOG+Q}} \tag{114}$$

confirms that both the MOG parameter and the quintessence field independently enlarge the photon capture cross-section, with their contributions being approximately additive.

V. CONCLUSION

In this work, we

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FIG. 10. Shadow boundary of the rotating MOG black hole surrounded by quintessence ($w_q = -2/3$, $M = 1$, observer at $\theta_o = \pi/2$). *Panel (a):* varying the MOG parameter $\alpha \in [0, 0.5]$ with $a = 0.5$ and $c = 0.005$; the shadow enlarges with α while retaining a nearly circular shape. *Panel (b):* varying the quintessence normalization $c \in [0, 0.02]$ with $a = 0.5$ and $\alpha = 0.3$; quintessence widens the shadow without altering the D-shape asymmetry. *Panel (c):* varying the spin $a \in \{0.1, 0.5, 0.7\}$ with $\alpha = 0.3$ and $c = 0.005$; increasing spin produces the characteristic D-shape, flattening the prograde side of the shadow due to frame-dragging. The maximum value $a = 0.7$ approaches the extremal spin $a_E \approx 0.75$ for these parameters (see Fig. **??**).

FIG. 11. Comparison of the shadow boundary for five spacetimes at $a = 0.5$, $w_q = -2/3$, and $\theta_o = \pi/2$: Kerr ($\alpha = 0$, $c = 0$), rotating MOG with $\alpha \in \{0.2, 0.4\}$ (no quintessence), and the full rotating MOG-quintessence geometry for the same α values with $c = 0.02$. The progressive enlargement of the shadow from Kerr outward demonstrates that the MOG coupling and quintessence field each independently increase the shadow radius, while the D-shape asymmetry is entirely governed by the spin a.